Lesson notes · for you

Attacks on passwords, and the defences

Cyber security · Level 3 · Lesson 3

Time	Materials
20 min	Paper and a pen (never write passwords on it); A phone or computer to change passwords, with an adult if needed

Learning goals

- Explain guessing, word lists, brute force and credential stuffing.
- Compare the possibilities in short and long passwords.
- Describe how websites, and you, defend accounts.

Activity: Password audit

1. List (4 min): Write every account you use: email, chats, games, school portal, social media. Never write a password on the page.
2. Mark (4 min): Circle the accounts that share a password with another one. Draw a line between each pair.
3. Rank (4 min): Number the accounts by how much harm a break-in would do. Email goes first, because it can reset all the others.
4. Fix (8 min): Change the top three to unique passphrases of random words, and turn on two-step sign-in where you can, with an adult if you need help.

The page lists account names only. If anyone asks to see your passwords for this activity, the answer is no.

Discussion

- Why is guessing on a website slow, but guessing after a leak fast?
- Why can a strong password still be dangerous?
- Which defence would you add first, and why?

Check yourself

- One attack on passwords + the defence that stops it.

Try it this week

List your accounts on paper, with no passwords on it. Mark every place you used the same password, then change those first, starting with your email, and turn on two-step sign-in where you can, with an adult's help if you need it.

Worksheet

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Name _____ Class _____ Date _____

1 **A gaming site you used years ago was hacked. You used the same password for your email. Why does it matter?**

- ◯ Attackers try the leaked password on your email too
- ◯ It doesn't — the gaming site is a different company

2 **Complete the sentences.**

Attackers try _____ passwords first. When a site's password file is _____, they can guess with no limit. Each extra random word _____ the possibilities.

Word bank: multiplies · rare · common · locked · halves · stolen

3 **Swapping letters for symbols, like P@ssw0rd, turns a weak password into a strong one.**

◯ True ◯ False

4 **Match each attack to the defence that stops it.**

Guessing common passwords	• • • •	Make it long: several random words
Trying every combination	• • • •	No names, birthdays or 123456
Credential stuffing	• • • •	Two-step sign-in
A password stolen by a fake page	• • • •	A different password for every account

5 **Which of these are good ways for a website to protect your password?**

- ☐ Refuse new passwords that appear in known leaks
- ☐ Store a fingerprint of the password, not the password
- ☐ Slow down or lock after too many wrong tries
- ☐ Force everyone to change it every month
- ☐ Email you your password when you forget it

✎ **Which of your accounts would hurt most if someone got into it? What protects it now, and what will you add?**

Check your answers

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1. **Attackers try the leaked password on your email too**

   It's a different company, but leaked email-and-password pairs are tried automatically on hundreds of other sites.

2. **Attackers try common passwords first. When a site's password file is stolen, they can guess with no limit. Each extra random word multiplies the possibilities.**

   Common passwords go first because people choose them.

3. **False**

   Attackers know those swaps (@ for a, 0 for o, 1 for i) and try them straight after the plain word.

4. **Guessing common passwords — No names, birthdays or 123456**
   Trying every combination — Make it long: several random words
   Credential stuffing — A different password for every account
   A password stolen by a fake page — Two-step sign-in

   Each attack has its own answer. Random words also avoid the common guesses, but only length beats trying every combination.

5. **Slow down or lock after too many wrong tries; Store a fingerprint of the password, not the password; Refuse new passwords that appear in known leaks**

   Locking slows guessing, a fingerprint means a stolen file doesn't show your password, and blocking leaked passwords stops easy guesses.